

# Problem Sheet 1 Solutions

1.

(a) We cannot increase the input to a sensor or instrument without limit and expect to get a useful output for all input values. Eventually, the system will “saturate” and will not be able to produce any output above some upper limit. The physical limit depends on the particular operating principle of the device.

Zero offset is the output given by a sensor for zero input. Ideally, we would like a nil zero-offset; however, it is a characteristic of almost all sensors that some small offset remains, no matter how much effort is put into the design and manufacture. Again, the zero offset is dependent on the physical operating principles.

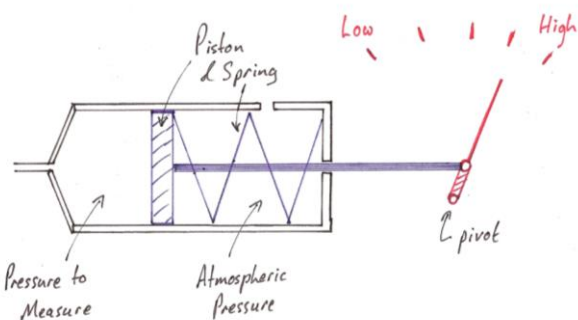
## (i) Mercury Thermometer

The mercury thermometer has a finite length of tube for the mercury to expand into. This sets a hard upper-limit on the temperature measurement. Increasing the temperature beyond this limit will not result in valid measurements – the thermometer is ‘saturated’. A zero offset can occur for several reasons: it may be that the reservoir may have been slightly over (or under) filled with mercury, in which case the intended zero-point will be higher (or lower) than the zero reading on the scale, or it is possible that the scale has been incorrectly aligned during manufacture.

## (ii) Mechanical Pressure Gauge

A mechanical pressure gauge uses the force applied to a piston, diaphragm or coiled tube to move an indicator needle. In the case of the design depicted in the figure, the pressure of the gas or liquid to be measured acts against a spring.

So long as the spring obeys Hooke’s law, the piston displacement is linearly proportional to the difference between the pressure to be measured and atmospheric pressure (note this is the definition of a pressure gauge – it measures pressure relative to atmospheric, not absolute pressure such as a barometer).



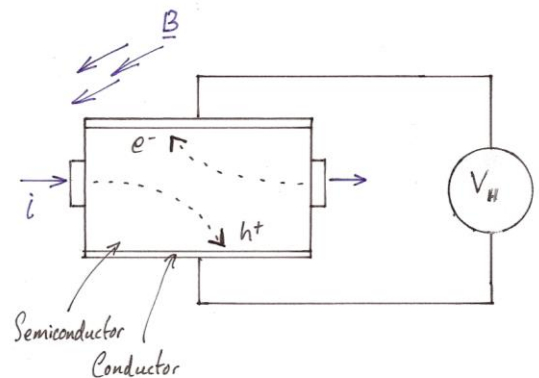
At very large differential pressures the spring will be compressed to its minimum length and the piston will no longer be able to move. The sensor will be ‘saturated’. A zero-offset error could have many causes, e.g.:

- Scale incorrectly aligned to the needle position;
- Friction between piston and tube resulting in the piston not reaching the true zero position (some residual compression remaining in the spring);
- Spring or rod manufactured to incorrect length.

## (iii) Hall Effect Probe

A Hall effect probe is typically implemented as a 4-terminal semiconductor device. A magnetic field perpendicular to the flow of current in the semiconductor deflects some charge carriers which then generate a transverse voltage which is a function of  $\underline{B}$ .

In the figure (right) the magnetic field is directed perpendicularly to the paper. As we increase  $\underline{B}$ , we eventually reach a situation where all the charge carriers are deflected to the plates and hence any further increase in  $\underline{B}$  would not increase the voltage – the probe is saturated.



The fact that the Hall sensor operates on very fundamental physical principles and has no moving parts makes it likely that the zero-offset should be very small. Its operation does require that the current distribution within the semiconductor, for zero magnetic field, be uniform. It is possible that manufacturing errors in the geometry or the purity of the semiconductor could contribute to a zero-offset. Another potential problem would be if there is any residual magnetism in the package of the device, e.g. due to any iron/steel or nickel. Of course the manufacturers will go to great lengths to avoid this, however some residual ‘magnetic contamination’ could cause an apparent offset.

The Hall sensor needs to be connected to an amplifier to amplify the very small Hall voltage to a level which can be read by a lab instrument such as a multimeter. For a given range of magnetic field input (both positive and negative), the output could be between +5V and -5V. Note, however, that some manufacturers add an electronic offset on purpose in order to make the output “unipolar”, e.g. 0 to 10V, with 0 to 5V representing negative magnetic field and 5V to 10V the positive input range. This is done to simplify the electronics somewhat (as we shall see later) and is an example of a *deliberate* offset; it is left to the user to remove this before interpreting the data.

(b) For each of these sensors, the zero-offset can be measured and then corrected for by subtraction of the offset from any subsequent measurements.

(i) For the thermometer the traditional method is to place it in a mixture of pure water and ice which is guaranteed to be at 0°C.

(ii) For the pressure gauge we simply disconnect it such that both sides are at atmospheric pressure.

(iii) For the Hall probe we need to expose it to zero magnetic field, which can be done by placing it inside a Helmholtz coil which has been driven with a current to oppose the Earth’s magnetic field. To know when we have achieved zero-field we would probably make use of a reference measurement device such as a high-accuracy magnetometer with an existing calibration traceable to national standards. First, we would establish the zero-field using our reference instrument, and then take the offset reading from our Hall sensor.

## 2.

(a) The coefficient of linear expansion  $\alpha$  is defined by

$$\alpha = \frac{1}{L} \frac{dL}{dT}$$

Let us say here that  $L$  represents the radius of the bulb containing the mercury, then the volume of mercury is  $\frac{4}{3}\pi L^3$ . The bulb has a fixed volume  $V$  but the mercury is free to expand up the bore, and the rate of change of mercury volume with temperature is

$$\begin{aligned} \frac{dV}{dT} &= \frac{dV}{dL} \cdot \frac{dL}{dT} \\ &= 4\pi L^2 \cdot \alpha L \\ &= 3\alpha V \end{aligned}$$

(b) This volume change equals the volume swept out as the mercury moves up the bore. Say the length of the mercury column in the bore is  $l$ , then the volume of the column is  $\pi r^2 l$

$$\begin{aligned} \pi r^2 \frac{dl}{dT} &= \frac{dV}{dT} \\ \frac{dl}{dT} &= \frac{3\alpha V}{\pi r^2} \end{aligned}$$

(c) Using  $\alpha = 1.82 \times 10^{-4} \text{ K}^{-1}$  and  $dl/dT = 10^{-3} \text{ m/}^\circ\text{C}$  and solving for  $r$  yields

$$r = 4.17 \times 10^{-4} \text{ m} \approx 0.4 \text{ mm.}$$

(d) The measurement range is 60°C, so we may choose to set the minimum at -10°C and the maximum at +50°C. The thermometer will saturate above this maximum. Using ethanol the temperature sensitivity increases by a factor of 6, therefore the range decreases by the same factor to 10°C and consequently the thermometer will be more easily saturated. Such high-sensitivity thermometers are manufactured for use in specialist applications such as chemistry or brewing where we need to make a high-resolution measurement over a narrow range of temperatures. The thermometer can be calibrated by testing at two known temperatures and inscribing the scale accordingly. Typically an ice/water mix is used to establish the 0°C point, which would establish and correct for any zero offset (which could be due to over-filling with mercury, for example). The temperature measured in pure boiling water, at standard atmospheric pressure, establishes 100°C.

### 3.

Refer to the AD590 datasheet pages 7-9. According to Table 4 of the datasheet, the time constant for the type 'H' package in stirred oil is 1.4 seconds. If  $T_0$  is the final temperature and  $T(t)$  is the temperature rise above ambient as a function of time  $t$  then we can write

$$T(t) = T_0 \left(1 - e^{-\frac{t}{\tau}}\right)$$

Where  $\tau$  is the time constant. For  $T = 0.865 T_0$

$$0.865 = 1 - e^{-\frac{t}{\tau}}$$

$$\frac{t}{\tau} = 2$$

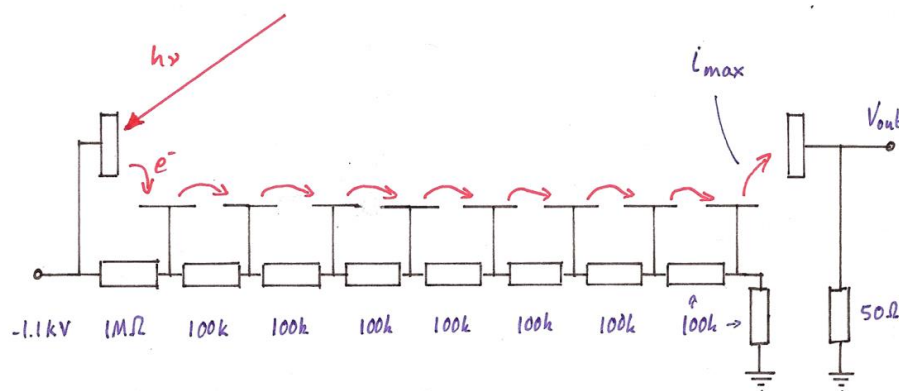
It therefore takes 2 time constants (= 2.8 s) to reach 86.5% of steady state. At  $t=14$  s,  $1 - e^{-\frac{t}{\tau}} = 0.9995$  which is an error of just 0.05°C at 100°C.

From figure 10 of the datasheet we can see that the uncorrected calibration error at 100°C is about 2°C. Figure 11 gives the non-linearity error as a function of temperature to be about 0.8°C. So, the error at  $t=14$  s is governed by these calibration effects.

Maximum error is 2.8°C and RMS Error is  $\sqrt{2^2 + 0.8^2} = 2.2^\circ\text{C}$ .

### 4.

(a) See figure below.



(b) First we calculate the electron gain of the tube. Each electron impact liberates 7 new electrons plus the impacting electron equals a total of 8. The total gain is therefore  $8^8$ . Factoring in quantum efficiency we obtain, *on average*,

$$0.35 \times 8^8 = 5.87 \times 10^6$$

This is the number of electrons at the anode per photon *incident* at the photocathode, which is a charge of  $9.4 \times 10^{-13}$  C.

[Note that, more commonly, you may want to calculate the gain of the PMT (i.e. the number of electrons arriving at the anode per photoelectron *emitted* from the photocathode), in which case you exclude the QE from the calculation.]

(c) When there is no light incident on the cathode (and no dark-current) all current flows through the resistor chain (conventional current flows from ground to the negative terminal of the HV supply, so here it is more useful to think of the current as the electron flow in the opposite direction). The current flowing through the resistor chain is

$$\frac{V}{R} = \frac{1.1 \text{ kV}}{1.8 \text{ M}\Omega} = 6.11 \times 10^{-4} \text{ A} = 6.11 \times 10^{-4} \text{ C/s.}$$

When light is incident on the cathode then electrons are liberated from the dynodes. The tube is saturated when all the current in the last resistor in the chain is diverted into the tube such that it returns to ground via the anode. This occurs when the incident light flux is

$$\frac{6.11 \times 10^{-4}}{9.4 \times 10^{-13}} = 6.50 \times 10^8 \text{ photons/s.}$$

The energy of a  $0.54 \mu\text{m}$  photon  $h\nu = 3.68 \times 10^{-19}$  J, so

$$\text{power} = 6.5 \times 10^8 \times 3.68 \times 10^{-19} = 2.4 \times 10^{-10} \text{ W.}$$

(d) Continuous saturation can result in destruction of the PMT. Hence, while powered the tube must be shielded from ambient light. If used in a location with strong magnetic fields, which can curve the electron paths, this can affect the gain of the tube by reducing the number of electrons reaching the anode.

## 5.

(a) Metric dynamic range = 600, Imperial dynamic range = 768. Imperial resolution is  $1/64^{\text{th}}$  of an inch  $\equiv 0.397 \text{ mm}$ , so the Imperial scale has the higher resolution.

(b) Regular micrometer

$$\text{dynamic range} = 20 \log \left( \frac{50}{0.01} \right) \cong 74 \text{ dB}$$

Vernier micrometer has a factor 10 greater resolution

$$\text{dynamic range} = 74 \text{ dB} + 20 \log 10 \cong 94 \text{ dB}$$

The equivalent metric ruler has length  $x$

$$\frac{x}{0.5} = \frac{50}{0.001}$$

$$x = 25000 \text{ mm or } 25 \text{ m}$$

(c) Note: you just need to calculate the means in order to see this. The standard deviations are interesting to know but actually it is obvious from the spread of the measurements which groups have the highest and lowest values. That said, both are below just for completeness (all in cm):

| Group | Mean   | Standard Deviation |
|-------|--------|--------------------|
| A     | 10.160 | 0.46               |
| B     | 10.146 | 0.08               |

|   |        |      |
|---|--------|------|
| C | 12.156 | 0.02 |
| D | 10.230 | 1.34 |
| E | 10.200 | 0.45 |

Before knowing the reference measurement:

1. Group B made the highest resolution measurements.  
Group E have the lowest resolution, measuring only to the nearest centimetre.
2. Group C has the most precise measurement since their standard deviation is the lowest.
3. Unknown, as we don't know the true value nor any reference measurement.
4. Group D have the worst repeatability; their standard deviation is the largest.

After knowing reference measurement

1. Group C. This statement is unchanged from (2) above since precision is determined by the measurement data only and knowing the (best estimate of) true value does not change this.
2. Group C has the largest error and the worst accuracy. Since their results are repeatable, however, we surmise that they have a precise measurement with small random error but a large systematic error.
3. Group A made the most accurate measurement. While their precision is quite poor, the mean of their results is closest to the true value.
4. Group D has the worst precision; again, same as (4) above.

Group A made measurements of relatively low resolution and precision but their mean is exactly the true value. We suspect that they got lucky! In the future, this may not happen again, therefore, we recommend they improve the precision of their measurements. If this is not possible then they should make many more measurements in the future such that their mean is more likely to be close to true value.

Group B has the highest resolution instrument and, apart from Group A, was closest to the true value. Their precision is surprisingly poor. We might guess that while they are using the best measurement instrument, they are not very good at using it properly. Better training could increase the repeatability of their measurements, and we might expect them to get the best result next time.

Group C has the best precision, comparable with the resolution of their instrument, so they are using their instrument well but it has a large systematic error. It needs to go back to the manufacturer for re-calibration!

Group D is quite close to true value but again this looks like a lucky shot. Their precision is appalling. They need to improve either equipment or their method.

Group E clearly has a really poor resolution instrument – something like a child's 10 cm ruler marked off in 1 cm graduations. Their only hope is to make a very large number of measurements.

## 6.

(a) The Thevenin equivalent circuit is an ideal voltage source  $V_{Th}$  in series with a resistance  $R_{Th}$ .

(b)  $V_{Th}$  is the open-circuit voltage across terminals AB. Since AB is open-circuit (zero load), no current flows in  $R_2$  hence by Ohm's law there is no potential difference across  $R_2$  and so the open-circuit voltage is the same as the voltage across  $R_3$  which is

$$V_{AB_{open}} = V_{Th} = \frac{V_1 R_3}{R_1 + R_3}$$

To find the Thevenin resistance we need to replace  $V_1$  with its circuit equivalent (a short-circuit, zero resistance) and evaluate the resistance at terminals AB, which is  $R_1$  in parallel with  $R_3$  and in series with  $R_2$

$$R_{Th} = R_2 + \left( \frac{1}{R_1} + \frac{1}{R_3} \right)^{-1} = R_2 + \frac{R_1 R_3}{R_1 + R_3}$$

Using the given values we get

$$V_{Th} = 7.5 \text{ V}$$

$$R_{Th} = 2.75 \text{ k}\Omega$$

The Thevenin equivalent circuit is an ideal voltage source of 7.5 V in series with a resistance of 2.75 k $\Omega$ .

(c) The Norton equivalent of the circuit is a current source  $i_N$  in parallel with the *same* resistance.

(d) We find the Norton current by evaluating that current which will produce the same open-circuit voltage as the Thevenin circuit:

$$i_N = \frac{7.5}{2750} = 2.73 \text{ mA}$$

## 7.

(a) At 100°C the LM35 looks like a voltage source of 1 V in series with a resistance of 0.1  $\Omega$ .

(b) If we denote the resistance of the meter by  $R$ , the voltage across the meter by  $V_R$  and the current drawn from the LM35 by  $i$ , then we can find  $i$  by dividing the source voltage by the sum of the two resistances, and get hence find  $V_R = iR$ .

| $R$         | $i$                         | $V_R$       |
|-------------|-----------------------------|-------------|
| 1 $\Omega$  | 0.909 A                     | 0.909 V     |
| 1k $\Omega$ | $9.999 \times 10^{-4}$ A    | 0.9999 V    |
| 1M $\Omega$ | $9.999999 \times 10^{-7}$ A | 0.9999999 V |

(c) The Norton equivalent circuit looks like a current source of 300  $\mu$ A in series with a resistance of 10 M $\Omega$ .

(d) We can evaluate the voltage across the parallel resistances (source and meter) and hence find the current which flows in the meter

| $R$         | $V_R$             | $i_R$             |
|-------------|-------------------|-------------------|
| 1 $\Omega$  | 299.99997 $\mu$ V | 299.99997 $\mu$ A |
| 1k $\Omega$ | 299.97 mV         | 299.97 $\mu$ A    |
| 1M $\Omega$ | 272.7 V           | 272.7 $\mu$ A     |

(e) The voltmeter should not load the sensor, i.e. it should not draw excess current which will drop a voltage across the sensor's internal resistance. A good voltmeter will have an input resistance of k $\Omega$ , and usually much higher. The ammeter should have a resistance as small as possible so as to not impede the flow of current. We wish that the voltage across the terminals, due to the presence of the meter, should stay as small as possible. A good ammeter will have an input resistance of less than a 1 Ohm.

## 8.

(a) A voltage source in series with resistance, as before. With no load connected we have the open-circuit voltage:

$$V_{Th} = \frac{VR_2}{R_1 + R_2}$$

The Thevenin resistance is the parallel combination of the 2 resistors

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2}$$

(b) When  $R_L$  is re-connected, the Thevenin circuit is 'loaded' and the measured voltage  $V_o$  becomes

$$V_o = \frac{V_{Th} R_L}{R_L + R_{Th}} = \frac{V R_2}{\left(\frac{R_1 R_2}{R_L}\right) + R_1 + R_2}$$

(c) As  $R_L \rightarrow \infty$  then  $V_o \rightarrow V_{Th}$ . This is expected since the ideal meter has infinite input impedance and would measure the Thevenin (open-circuit) voltage of the source.